Equipping School Psychologists with Tools to Promote Early Math Skills

INTRODUCTION

- Early math skills are the best predictor of school readiness and academic success (Duncan et al., 2007, Claessens & Engel, 2013), but little time is spent on math in preschool (Piasta, Pelatti, & Miller, 2014).
- Preschool teachers need to know effective early math interventions.
- The purpose of this study is to synthesize research on early math interventions.

METHOD

- The initial search yield was 3,934. After screening, 16 articles remained.
- Articles were analyzed for quality using the Council of Exceptional Children's (CEC) Standards for Classifying the Evidence Base of Practices in Special Education (Cook et al., 2014).

CONCLUSIONS

- The majority of effective early math interventions target patterning (7) and numeracy (6) skills.
- Patterning interventions included oddity instruction, advance organizer lessons, explanations, and block play.
- Numeracy interventions included instructional guidance, creating opportunities and prompting, explicit instruction in counting, direct instruction flashcards, modeling procedures, and playing a board game.
- Some interventions focused on geometry (4) by creating opportunities and prompting, , direct instruction flashcards, block play, and modeling.
- Two interventions focused on overall math skills by incorporating math activities into class routines and reading.
- Finally, there were no activities found to target early measurement skills.

There is a lack of research on effective preschool math interventions targeting measurement skills, general math skills and geometry.

Most research targets patterning and numeracy skills.

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Table 1

Inclusion and Exclusion Criteria

Area of the Study	Inclusion Criteria	Exclusion Criteria
Design	RCTs, experimental designs, single-case designs, or quasi-experimental design	Correlational, qualitative or case study
Publication Type	Published, peer-reviewed literature	
Geography	United States	
Age Range	Early childhood or preschool	Kindergarten and older
Setting	Any early childhood learning center	Home-based childcare
Outcome	math skills at the end of intervention or the beginning of kindergarten as measured by direct assessment, informal or formal rating scales, or teacher report	
Intervention Agent	Any teacher or other adult	Parents and guardians
Intervention	Math intervention	Formal curriculum; intervention that requires digital technology
Language	English	

Appendix A.

Coding Procedures

Area of the Study	Components Coded
Design	RCT, experimental design, single-case design, or quasi-experimental design Demographics of participants Number of participants
Setting	Type of setting Geographical location
Intervention or Strategy	Name of intervention Intervention agent Frequency of implementation Duration of intervention Length of individual sessions Number of students receiving intervention at once (individual, small group, or whole class)
Outcome	Effect size measured Qualitative report of outcomes Measure used

Appendix B.

PRISMA Chart of Inclusion Process

